In [1]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("ecommerce_dirty_dataset_15_records.csv")

df.head()
```

Out[1]: Columns 1–8 of 10

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
0	1	03-01-2026	Wireless Mouse	Electronics	2	599	asha@example.com
1	2	04-01-2026	USB-C Cable	electronics	3	199	ravi@example.com
2	3	05-01-2026	Bluetooth Speaker	Audio	1	2499	meena@example.com
3	4	06-01-2026	Laptop Stand	Accessories	NaN	899	kiran@example.com
4	5	07-01-2026	Wireless Keyboard	Electronics	two	1299	divya@example.com

Columns 9–10 of 10 · first column repeated

	Payment_Status	Rating
0	Paid	5.0
1	paid	4.0
2	Completed	5.0
3	Paid	4.0
4	Pending	3.0

In [2]:
```
df.isnull().sum()
```

Out[2]:
```
Order_ID          0
Order_Date        0
Product_Title     1
Category          0
Quantity          1
Unit_Price        1
Customer_Email    1
Payment_Status    0
Rating            1
dtype: int64
```

In [3]:
```
df.duplicated().sum()
```

Out[3]:
```
np.int64(1)
```

In [4]:
```
df[df.duplicated()]
```

Out[4]: Columns 1–8 of 10

	Order_ID	Order_Date	Product_Title	Category	Quantity	Unit_Price	Customer_Email
15	2	04-01-2026	USB-C Cable	electronics	3	199	ravi@example.com

Columns 9–10 of 10 · first column repeated

	Payment_Status	Rating
15	paid	4.0

In [5]:
```
df = df.drop_duplicates()
```

In [6]:
```
df["Product_Title"] = df["Product_Title"].str.strip()
```

In [7]:
```
df["Quantity"] = pd.to_numeric(
    df["Quantity"], errors="coerce"
)
```

In [9]:
```
df["Quantity"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Quantity
Non-Null Count  Dtype
--------------  -----
13 non-null     float64
dtypes: float64(1)
memory usage: 252.0 bytes
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        15 non-null     int64
 1   Order_Date      15 non-null     str
 2   Product_Title   14 non-null     str
 3   Category        15 non-null     str
 4   Quantity        13 non-null     float64
 5   Unit_Price      14 non-null     str
 6   Customer_Email  14 non-null     str
 7   Payment_Status  15 non-null     str
 8   Rating          14 non-null     float64
dtypes: float64(2), int64(1), str(6)
memory usage: 1.2 KB
```

In [11]:
```
df["Unit_Price"] = pd.to_numeric(
    df["Unit_Price"], errors="coerce"
)
```

In [13]:
```
df["Unit_Price"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Unit_Price
Non-Null Count  Dtype
--------------  -----
13 non-null     float64
dtypes: float64(1)
memory usage: 252.0 bytes
```

In [14]:
```
df["Order_Date"] = pd.to_datetime(
    df["Order_Date"], errors="coerce", dayfirst=True
)
```

In [15]:
```
df["Order_Date"].info()
```

```
<class 'pandas.Series'>
RangeIndex: 15 entries, 0 to 14
Series name: Order_Date
Non-Null Count  Dtype
--------------  -----
14 non-null     datetime64[us]
dtypes: datetime64[us](1)
memory usage: 252.0 bytes
```

In [16]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        15 non-null     int64
 1   Order_Date      14 non-null     datetime64[us]
 2   Product_Title   14 non-null     str
 3   Category        15 non-null     str
 4   Quantity        13 non-null     float64
 5   Unit_Price      13 non-null     float64
 6   Customer_Email  14 non-null     str
 7   Payment_Status  15 non-null     str
 8   Rating          14 non-null     float64
dtypes: datetime64[us](1), float64(3), int64(1), str(4)
memory usage: 1.2 KB
```

In [17]:
```
df.shape
```

Out[17]:
```
(15, 9)
```

In [18]:
```
df.dtypes
```

Out[18]:
```
Order_ID                    int64
Order_Date         datetime64[us]
Product_Title                 str
Category                      str
Quantity                  float64
Unit_Price                float64
Customer_Email                str
Payment_Status                str
Rating                    float64
dtype: object
```

In [19]:
```
df.columns
```

Out[19]:
```
Index(['Order_ID', 'Order_Date', 'Product_Title', 'Category', 'Quantity',
       'Unit_Price', 'Customer_Email', 'Payment_Status', 'Rating'],
      dtype='str')
```